Predicting Tomorrow's Recovery: A WHOOP Personal Data Project

Executive Summary

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Dataset: 237 days of personal WHOOP data (Sept 17, 2025 to May 12, 2026)

Model	RMSE	MAE	R^2
Mean baseline (DummyRegressor)	24.46	20.04	~0.00
Pure autoregression, AR(7)	25.01	20.79	-0.05
KNN regressor, K = 3	23.39	17.98	0.09
Ridge AR + exogenous variables	**20.05**	**15.03**	**0.33**

A logistic regression model predicting WHOOP's Red / Yellow / Green recovery zones reached 54.3% test accuracy, compared to a 41.3% always-Yellow baseline. KMeans clustering also produced a surprisingly clean separation between "good," "average," and "bad" physiological days that lined up closely with WHOOP's own recovery categories and with certain lifestyle habits.

Introduction

This project uses my personal WHOOP wearable data, covering around eight months of physiology, sleep, workouts, and lifestyle information, to answer a simple question: Can tomorrow's Recovery score be predicted using today's data?

Recovery is WHOOP's headline readiness metric, scored from 0 to 100, and is meant to summarize how prepared the body is for physical strain. Since it is the metric I personally pay the most attention to before deciding how hard to train, I wanted to see whether there was a meaningful predictive signal behind it.

The project combines three different approaches from the Data Bootcamp toolkit. First, it builds regression models that try to predict the next day's Recovery score using autoregressive lag features and physiological variables. Second, it reframes the problem as a classification task using WHOOP's Red, Yellow, and Green recovery zones. Finally, it applies unsupervised clustering to identify broader "good day versus bad day" patterns in the data.

Data Description

WHOOP exports four main CSV files. The most important is physiological_cycles.csv, which contains one row per day with Recovery %, heart-rate variability (HRV), resting heart rate (RHR), skin temperature, blood oxygen, Day Strain, calories burned, heart-rate metrics, and detailed sleep statistics such as REM, deep sleep, sleep debt, efficiency, and consistency.

The dataset also includes:

- workouts.csv for workout-level information
- sleeps.csv for sleep event details
- journal_entries.csv for daily yes/no lifestyle questions

The journal data included variables such as caffeine intake, alcohol consumption, late eating, magnesium use, protein tracking, screens in bed, sunlight exposure after waking, and calorie tracking.

These datasets were merged by date inside src/features.py. Additional features were then engineered, including:

- Recovery lag variables from 1 to 7 days
- HRV, RHR, strain, and sleep lag variables
- Rolling 7-day workout averages
- Sleep timing variables
- Day-of-week and weekend indicators

After preprocessing and feature engineering, the final modeling dataset contained 226 usable days and 64 total features.

Methodology and Train/Test Split

Because this project focuses on prediction, the train/test split was one of the most important methodological decisions in the analysis.

I used scikit-learn's train_test_split with shuffle=False, meaning the earliest 80% of observations formed the training set while the most recent 20% formed the test set. This was necessary because the model relies heavily on lagged variables such as previous Recovery scores and previous HRV measurements. Randomly shuffling the data would have introduced data leakage by placing adjacent days on opposite sides of the split.

The training period covered Sept. 19, 2025, through March 25, 2026, while the test period covered March 26 through May 11, 2026.

To maintain time-series discipline:

- Exploratory data analysis only used the training set
- KNN hyperparameters were tuned using TimeSeriesSplit
- Test predictions were generated only once at the end of modeling

I also ran a sensitivity check using a random shuffled split averaged over 50 seeds. The results were noticeably different, reinforcing that only the time-aware split produces valid performance estimates in this setting.

Models and Methods

The project used pandas, NumPy, matplotlib, seaborn, scikit-learn, and statsmodels.

Five predictive approaches were compared:

Baseline Model

A DummyRegressor(strategy='mean') served as the baseline benchmark.

Pure Autoregression

Autoregressive models were implemented using both statsmodels.AutoReg and sklearn.LinearRegression on lag features. The two implementations produced identical coefficients, confirming consistency between frameworks.

Before fitting the AR model, standard diagnostics were performed:

- The Augmented Dickey-Fuller test rejected the null of a unit root, indicating stationarity
- The ACF decayed gradually
- The PACF showed noticeable spikes at lags 1 and 5

Although AIC selected an AR(7) specification, only lag 1 and lag 5 were individually significant in the final model. Residual autocorrelation checks showed that the autoregressive model had already captured most of the meaningful linear memory present in the series.

Ridge Regression with Exogenous Variables

The strongest model combined autoregressive features with physiological variables, sleep metrics, workout statistics, and journal variables. Ridge regression was used because many sleep metrics were highly correlated with one another.

KNN Regression

A KNN regressor with K = 3 was selected using 5-fold TimeSeriesSplit cross-validation.

Logistic Regression and KMeans

The classification model predicted WHOOP recovery zones:

- Red: below 34%
- Yellow: 34% to 66%
- Green: above 67%

KMeans clustering was also used to group days into broader physiological profiles.

Results and Interpretation

One of the most interesting findings was that Recovery turned out to be genuinely difficult to predict.

The mean baseline already performed reasonably well, and the pure autoregressive model actually performed slightly worse. This suggests that WHOOP Recovery is more mean-reverting than strongly persistent over time. While the autoregressive model captured most of the available linear memory in the series, there simply was not that much memory to capture.

The Ridge model performed substantially better:

Model	RMSE	MAE	R^2
Baseline	24.46	20.04	~0.00
AR(7)	25.01	20.79	-0.05
KNN	23.39	17.98	0.09
Ridge AR + Exogenous	20.05	15.03	0.33

The Ridge model reduced prediction error by roughly five percentage points and explained about one-third of the variance in Recovery scores.

The classification results told a similar story. Logistic regression achieved 54.3% accuracy compared to a 41.3% always-Yellow baseline. Most classification mistakes were only off by one category rather than completely incorrect.

The clustering analysis also produced intuitive patterns. “Bad day” clusters tended to feature:

- Lower HRV
- Higher resting heart rate
- Less sleep
- More alcohol and late-night eating entries

Meanwhile, “good day” clusters were associated with:

- Higher sleep quality
- Sunlight exposure after waking
- Consistent protein tracking

Conclusion and Next Steps

The main takeaway from this project is that Recovery is noisy and only moderately predictable from one day to the next. Autoregressive information alone contributes very little predictive power. Most of the signal comes from broader physiological and behavioral variables such as HRV, sleep performance, resting heart rate, and recent training load.

Still, the final Ridge model achieved an out-of-sample R^2 of 0.33 on unseen data, which suggests there is a real and reproducible signal behind WHOOP Recovery.

Several future improvements could strengthen the project further:

- Expanding the dataset to a longer time horizon
- Adding more journal variables, such as stress, hydration, or meditation
- Testing stronger non-linear models like gradient boosting
- Exploring Hidden Markov Models for persistent recovery "regimes"

Overall, the project demonstrates that wearable-device data contains meaningful predictive information, even if human recovery remains partly unpredictable by nature.

The point of this project is genuinely forecasting future Recovery, not just evaluating a model on data where the answers are already known. WHOOP's daily Recovery score, ranging from 0 to 100%, is the metric I personally check every morning to decide how hard to train. Because of that, the real objective of the project is not simply achieving a strong test-set score, but building a model that can actually estimate tomorrow's Recovery before tomorrow happens.

The pipeline generates three progressively more realistic forms of prediction.

First, the model predicts next-day Recovery across the held-out 46-day test period from March 26, 2026 through May 11, 2026. On this standard static evaluation, the Ridge autoregression plus exogenous variables model achieved an RMSE of 20.05 and an R^2 of 0.33. In practical terms, predictions are typically within about 15 percentage points of the true Recovery score and explain roughly one-third of the observed day-to-day variation. This serves as the project's clean, out-of-sample benchmark.

Second, the same model is evaluated in a walk-forward forecasting simulation. For each day in the test period, the model is retrained using all prior observations and then used to forecast the following day, mirroring how a real deployed application would behave in practice. Under this framework, performance improves slightly to RMSE 19.26 and R^2 0.38, likely because the model continually incorporates the most recent physiological information and behavioral context.

Finally, and most importantly, the project produces a true forward-looking prediction. Using the latest available WHOOP cycle from May 12, 2026, the fully trained model forecasts a Recovery score of 58.4% for May 13, 2026, corresponding to the Yellow Recovery zone. Unlike the held-out test predictions, this value does not already exist anywhere in the dataset. Re-running the notebook with a newly appended daily cycle automatically generates a fresh prediction for the following day.